



SYDNEY BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 2

Trial Higher School Certificate 2001

Time Allowed: 3 hours (plus 5 minutes reading time)

Total Marks: 120

Examiner: Mr R Dowdell, Mr PS Parker

INSTRUCTIONS:

- Attempt *all* questions.
- *All* questions are of equal value.
- All necessary working should be shown in every question. Full marks may not be awarded if work is careless or badly arranged.
- Standard integrals are provided on the last page. Approved calculators may be used.
- Return your answers in 8 booklets, 1 for each question. Each booklet must show your name.
- If required, additional Writing Booklets may be obtained from the Examination Supervisor upon request.

NOTE: This is a trial paper only and does not necessarily reflect the content or format of the final Higher School Certificate Examination Paper for this subject.

Question 1:

Marks

(a) Evaluate $\int_0^{\frac{3}{2}} \frac{dx}{\sqrt{9-x^2}}$

2

(b) Find $\int x^3 e^{x^4+7} dx$

2

(c)

(i) Express $\frac{x^2+x+2}{(x^2+1)(x+1)}$ in the form $\frac{Ax+B}{x^2+1} + \frac{C}{x+1}$, where A , B and C are constants.

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(ii) Hence find $\int \frac{x^2+x+2}{(x^2+1)(x+1)} dx$.

(d)

Using integration by parts or otherwise, evaluate $\int_0^{\frac{1}{2}} \sin^{-1} x dx$

3

(e)

By using the substitution $x = \pi - y$, or otherwise, evaluate $\int_0^{\pi} x \sin^3 x dx$

5

Question 2: START A NEW BOOKLET

Marks

(a) $\frac{4+3i}{1+\sqrt{2}i} = a+ib$, for a, b real.

2

Find the exact values of a and b .

(b) Given $z = 1 - \sqrt{3}i$,

3

(i) show that z^2 is a real multiple of $\frac{1}{z}$;

(ii) plot $z, z^2, \frac{1}{z}$ on an Argand diagram.

(c) Sketch the region represented by

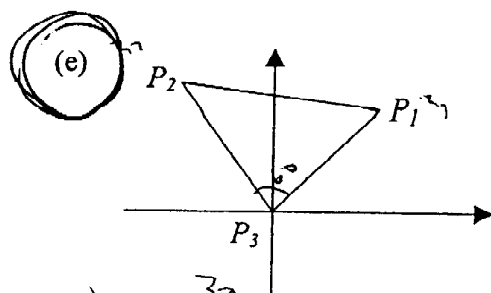
2

$$|z| \leq 4 \text{ and } \frac{\pi}{3} < \arg z \leq \frac{2\pi}{3}.$$

(d) (i) Show that $\frac{(1+i\sqrt{3})^6}{(\sqrt{3}-i)^k} = 2^{6-k} \operatorname{cis}\left(\frac{k\pi}{6}\right)$.

4

(ii) For what values of k is $\frac{(1+i\sqrt{3})^6}{(\sqrt{3}-i)^k}$ purely imaginary?



The points P_1, P_2 and P_3 represent the complex numbers z_1, z_2 and z_3 respectively. (NOTE: $z_3 = 0$.)

4

If P_1, P_2 and P_3 are the vertices of an equilateral triangle, show that

$$\frac{z_2}{z_1} = \frac{1+i\sqrt{3}}{2} \text{ and deduce that } z_1^2 + z_2^2 = z_1 z_2.$$

(ii) Deduce that if z_1, z_2 and z_3 are ANY three complex numbers at the vertices of an equilateral triangle then

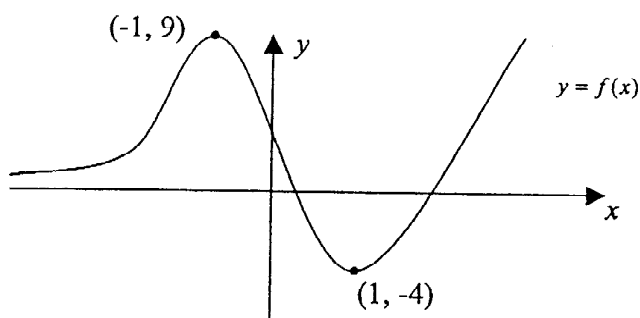
$$z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$$

Question 3: START A NEW BOOKLET

Marks

- (a) If the curve below represents $y = f(x)$,

12



make neat sketches, on separate axes, of

- (i) $y = (f(x))^2$
- (ii) $y = \frac{1}{f(x)}$
- (iii) $y = |f(x)|$
- (iv) $y = f(|x|)$
- (v) $y^2 = f(x)$
- (vi) $y = f'(x)$

(b)

Two sides of a triangle are in the ratio 3:1 and the angles opposite these sides differ by $\frac{\pi}{6}$. Show that the smaller of the two angles is $\tan^{-1}\left(\frac{1}{6-\sqrt{3}}\right)$.

3

Question 4: START A NEW BOOKLET

Marks

- (a) $1+i$ and $3-i$ are zeroes of a real, monic polynomial, $p(x)$, of degree 4. 3

(i) Express $p(x)$ as a product of two real quadratic factors.

(ii) Explain briefly why the polynomial $p(x)$ cannot take negative values.

- (b) $x^3 + 3px + q = 0$ has a double root of $x = k$. 4

(i) Show that $p = -k^2$.

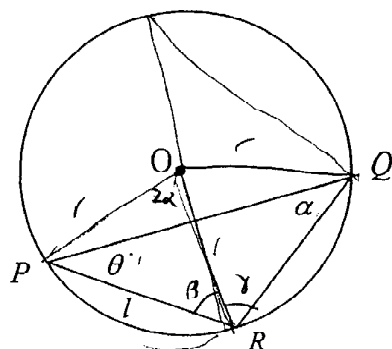
(ii) Show that $4p^3 + q^2 = 0$.

(iii) Hence factorise $x^3 - 6ix + 4 - 4i$ into linear factors, given that it has a repeated factor.

- (c) Consider $f(x) = x^3 + 9x + 26$ and $g(x) = x^2 + 26x - 27$. 3

(i) Verify that $f\left(x - \frac{3}{x}\right) = \frac{g(x^3)}{x^3}$.

(ii) Hence solve $f(x) = 0$.



ΔPQR is a triangle inscribed in a circle of radius r . PR has length l , and $\angle PQR = \alpha$

(i) Show that $l = 2r \sin \alpha$.

(ii) If $\angle QPR = \theta$, show that the area of ΔPQR is $r^2 \sin \alpha (\cos \alpha - \cos(2\theta + \alpha))$

(iii) If $PQ = QR$, what is the area of ΔPQR in terms of r and α ?

Question 5: START A NEW BOOKLET

Marks

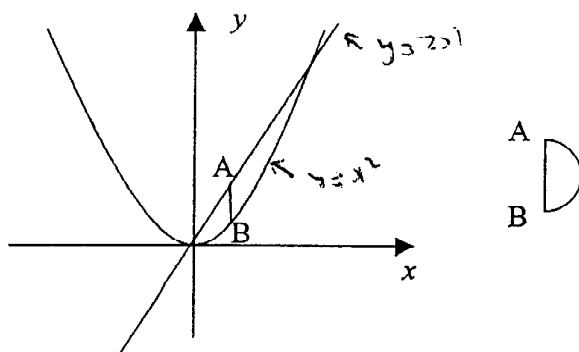
- (a) A mass of m kilograms falls from rest. It experiences resistance during its fall equal to mkv where v is its speed in metres per second and k is a positive constant. Let x be the distance in metres of the mass from its starting point measured positively as it falls and t be the time in seconds. 8
- (i) Show that the equation of motion of the mass is $\ddot{x} = g - kv$ where g is the acceleration due to gravity.
 - (ii) Show that the terminal velocity is $\frac{g}{k}$.
 - (iii) Find v as a function of t .
 - (iv) Find x as a function of t .
- (b) (i) In how many ways can 10 students be grouped into two teams of 5 to play a game of basketball? 2
- (ii) Two of the 10 students are twins. If the teams are formed at random, what is the probability that the twins play on the same team?
- (c) A group of men and women is seated randomly around a circular table. What is the probability that none of the men are sitting next to each other if there are 5
- (i) 3 men and 2 women;
 - (ii) 2 men and 3 women;
 - (iii) n men and $n + 1$ women?

Question 6: START A NEW BOOKLET

Marks

- (a) The base of a solid is the region enclosed by $y = 2x$ and $y = x^2$. Cross sections taken perpendicular to the x axis are semicircles with the diameter in the base of the solid (as indicated the diameter AB of the semicircle is perpendicular to the x axis; the semicircle is perpendicular to the xy plane).

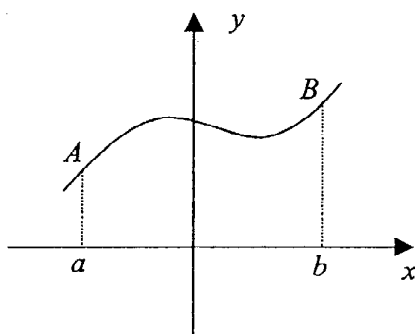
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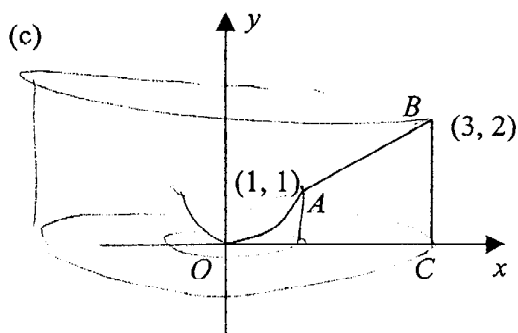
Find the volume of the solid.

- (b) The length of the arc AB on the curve $y = f(x)$ between $x = a$ and $x = b$ is given by $l = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$.

4



Find the length of the arc on $y = x^{\frac{3}{2}}$ between $x = 0$ and $x = 4$.



OA is an arc of the parabola $y = x^2$. The region $OABC$ is rotated about the y axis forming a bowl. By using cylindrical shells determine the holding capacity of the bowl.

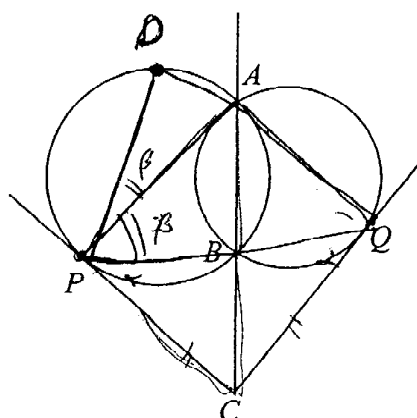
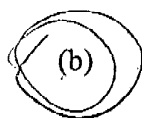
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Question 7: START A NEW BOOKLET

Marks

- (a) Find the value of a given that $\left(\sqrt{x} + \frac{a}{x}\right)^{10}$ has 13440 as coefficient of x^{-4} .

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Two circles intersect at A and B . AB is produced to a point C , such that CP and CQ are tangents to the circles as shown and PBQ is a straight line.

NOTE: The diagram is not drawn to scale.

- Express CP in terms of CB and CA , and hence prove that $CP = CQ$.
- Show that A, P, C and Q are concyclic.
- Let QA produced meet the larger circle at D . Show that PB bisects $\angle CPD$.

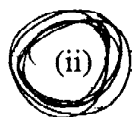
(c) Let $T(m, y) = \frac{{}^m C_0}{y} - \frac{{}^m C_1}{y+1} + \frac{{}^m C_2}{y+2} - \dots + (-1)^m \frac{{}^m C_m}{y+m}$.

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- If it is given that $T(k, x) = \frac{k!}{x(x+1)(x+2)\dots(x+k)}$ for a particular value of k , show that

$$T(k, x) - T(k, x+1) = T(k+1, x)$$

Use
the
induction



Hence prove, using Mathematical Induction or otherwise, that for $n \geq 1$

$$T(n, x) = \frac{{}^n C_0}{x} - \frac{{}^n C_1}{x+1} + \frac{{}^n C_2}{x+2} - \dots + (-1)^n \frac{{}^n C_n}{x+n} = \frac{n!}{x(x+1)(x+2)\dots(x+n)}$$

(NOTE: you may use without proof the result ${}^{m+1}C_r = {}^m C_r + {}^m C_{r-1}$)

- Hence prove that

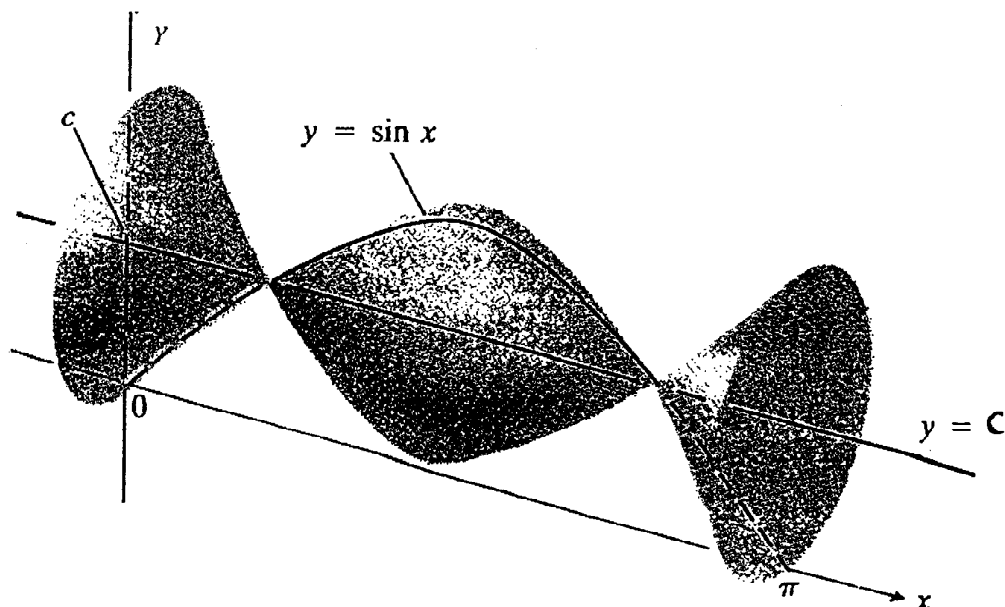
$$\frac{{}^n C_0}{1} - \frac{{}^n C_1}{3} + \frac{{}^n C_2}{5} - \dots + (-1)^n \frac{{}^n C_n}{2n+1} = \frac{2^n n!}{1.3.5\dots(2n+1)}$$

Question 8: START A NEW BOOKLET

Marks

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(a)



The arch $y = \sin x$, $0 \leq x \leq \pi$ is revolved around the line $y = c$ to generate the solid shown. Find the value of c that minimises the volume.

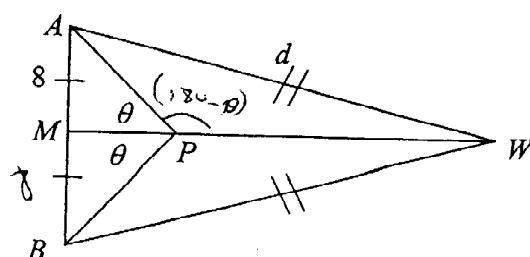
Question 8 is continued on Page 10

- (b) (i) Let $f(\theta) = \frac{2 - \cos \theta}{\sin \theta}$, $0 < \theta < \frac{\pi}{2}$.

Show that $f'(\theta) = \frac{1 - 2 \cos \theta}{\sin^2 \theta}$.

Find the minimum value of $f(\theta)$.

- (ii) Two towns A and B are 16km apart, and each at a distance of d km from a water well at W . Let M be the midpoint of AB , P be a point on the line segment MW , and $\theta = \angle APM = \angle BPM$. The two towns are to be supplied with water from W , via three straight water pipes: PW , PA and PB as shown below.



Show that the total length of the water pipe L is given

by $L = 8f(\theta) + \sqrt{d^2 - 64}$, when $\frac{8}{d} \leq \sin \theta \leq 1$, where $f(\theta)$ is given in part (i).

- (iii) If $d = 20$, find the length of MP when L is minimum, and the minimum value of L .
Show that this minimum value of L is less than the sum of any pair of sides of $\triangle ABW$.
- (iv) If $d = 9$, show that the minimum value of L cannot be found by using the same methods as used in part (iii). Explain briefly how to find the minimum value of L in this case. (Hint: Draw a diagram which illustrates this situation.)

END OF PAPER



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2001

TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics Extension 2

Sample Solutions

Question 1.

$$(a) \int_0^{\frac{\pi}{2}} \frac{dx}{\sqrt{1-x^2}} = \left[\sin^{-1} x \right]_0^{\frac{\pi}{2}} \\ \sin^{-1} \frac{\pi}{2} - \sin^{-1} 0 \\ = \left(\frac{\pi}{2} \right) \quad \checkmark \quad \checkmark \quad (1)$$

$$(b) \int x^3 e^{x^2+7} dx = \left(\frac{1}{4} e^{x^2+7} + C \right) \quad \checkmark \quad \checkmark \quad (2)$$

$$(c) (i). \frac{x^r + x + 2}{(x^r+1)(x+1)} = \frac{Ax+B}{x^r+1} + \frac{C}{x+1} \Rightarrow x^r + x + 2 = (Ax+B)(x+1) + C(x^r+1) \\ = Ax^r + Ax + Bx + B + Cx^r + C \\ = (A+C)x^r + (A+B)x + B+C$$

$$\therefore \int \frac{x^r + x + 2}{(x^r+1)(x+1)} dx = \int \left(\frac{1}{x^r+1} + \frac{1}{x+1} \right) dx: \quad \left. \begin{array}{l} A+C=1 \\ A+B=1 \end{array} \right\} \begin{array}{l} B-C=0 \text{ --- (1)} \\ B=C=2 \text{ --- (2)} \end{array} \\ \left. \begin{array}{l} 0+2 \\ 2B=2 \\ B=1 \\ C=1 \\ A=0 \end{array} \right\} \quad \checkmark$$

$$(d) \int_0^{\frac{1}{2}} x \sin^2 x dx = \int_0^{\frac{1}{2}} \frac{d}{dx} (x) \cdot \sin^2 x dx \\ = \left[x \sin^2 x \right]_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} x \cdot \frac{1}{1-x^2} dx \quad \checkmark \\ = \frac{1}{2} \times \frac{\pi}{6} + \left[\sqrt{1-x^2} \right]_0^{\frac{1}{2}} \quad \checkmark \\ = \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1 \quad \checkmark$$

$$(e) I = \int_0^{\pi} x \sin^3 x dx \\ = - \int_0^{\pi} (\pi-y) \cdot \sin^3(\pi-y) dy \\ = - \int_0^{\pi} (\pi-y) \sin^3 y dy \\ = \int_0^{\pi} (\pi-y) \sin^3 y dy \\ = \int_0^{\pi} (\pi-y) \sin^3 y dy \\ I = \int_0^{\pi} \pi \sin^3 y dy - \int_0^{\pi} y \sin^3 y dy \\ \therefore 2I = \pi \int_0^{\pi} \sin^3 y dy - \int_0^{\pi} y \sin^3 y dy \quad \text{let } u = \pi - y \\ du = -dy \\ = \pi \int_0^{\pi} (\pi-u) \sin^3 u du \\ = \pi \int_0^{\pi} (\pi-u) \sin^3 u du = 2\pi \int_0^{\pi} (\pi-u) \sin^3 u du \quad \left[\frac{u}{3} - \frac{1}{3} \right]_0^{\pi} \\ \therefore I = \frac{\pi}{3} \left[\frac{u}{3} - \frac{1}{3} \right]_0^{\pi} = \frac{\pi}{3} \left[\frac{\pi}{3} - \frac{1}{3} \right] = \frac{\pi}{3} \left[\frac{\pi-1}{3} \right] = \frac{\pi(\pi-1)}{9}$$

QUESTION 2 (a) $\frac{4+3i}{1+i\sqrt{3}} = \frac{4+3i}{1+i\sqrt{3}} \times \frac{1-i\sqrt{3}}{1-i\sqrt{3}}$

$$= \frac{4-4i\sqrt{3}+3i+3i\sqrt{3}}{3}$$

$$= \frac{4+3i\sqrt{3}-4i+3i\sqrt{3}}{3}$$

$\therefore a = \frac{4+3i\sqrt{3}}{3}, b = \frac{3-i\sqrt{3}}{3}$ (2)

(b) $z = 1-i\sqrt{3}$ $\frac{1}{z} = \frac{1}{1-i\sqrt{3}} \times \frac{1+i\sqrt{3}}{1+i\sqrt{3}}$

$$z^2 = 2-2i\sqrt{3} = -2(1+i\sqrt{3})$$

$$= \frac{1+i\sqrt{3}}{4} = \frac{1}{4}(1+i\sqrt{3})$$

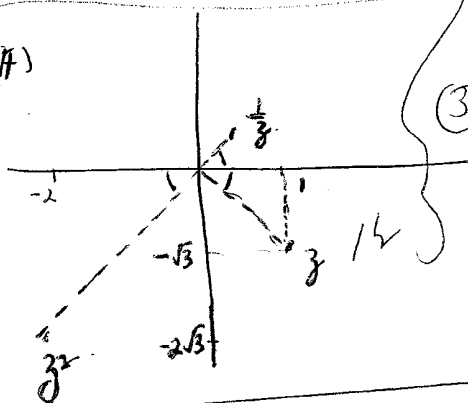
$z = 2\cos\frac{\pi}{3}$

$z^2 = 4\cos\frac{2\pi}{3}$

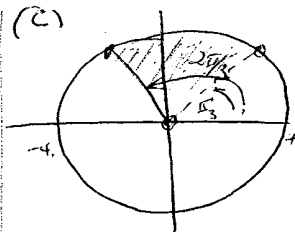
$\frac{1}{z} = \frac{1}{2}\cos\frac{\pi}{3}$

$z^2 = -8 \times \frac{1}{8}$ OR SHOW $z^2 = -8$

(7)



(8)



(2)

(i) $\frac{(1+i\sqrt{3})^6}{(\sqrt{3}-i)^6} = \frac{(2\cos\frac{\pi}{3})^6}{(2\cos(-\frac{\pi}{6}))^6}$

$$= 2^{6-k} \cos(2\pi + \frac{k\pi}{6})$$

$$= 2^{6-k} \cos\frac{k\pi}{6}$$

ii) $\frac{z}{\text{imag.}} \cos\frac{k\pi}{6} = 0$

$$\frac{k\pi}{6} = 2n\pi \pm \frac{\pi}{2}$$

$k = 3(4n+1)$ $n \in \mathbb{Z}$

$k = \pm 3, \text{etc}$

$12n \pm 3 \Rightarrow \pm 3, \pm 9, \pm 15, \dots$

$3, 9, 15, 21, \dots$

$-3, -9, -15, \dots$

$\pm 3, \pm 9, \pm 15, \pm 21, \dots$

$0-k (6n+3) n \in \mathbb{Z}$

(e). (i) now $z_2 = \omega_{\frac{\pi}{3}} \cdot z_1$

$$\therefore \frac{z_2}{z_1} = \omega_{\frac{\pi}{3}} = \frac{1+i\sqrt{3}}{2} \quad \text{--- 1}$$

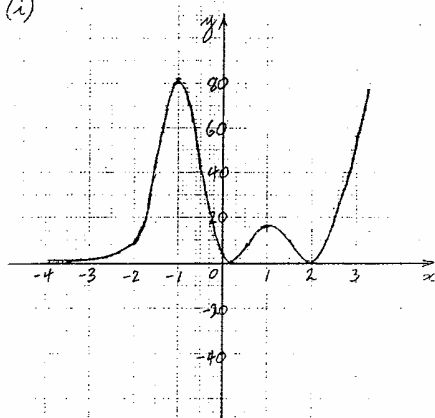
$$\begin{aligned} \therefore z_1^2 + z_2^2 &= z_1^2 + (\omega_{\frac{\pi}{3}} z_1)^2 \\ &= z_1^2 (1 + \omega_{\frac{\pi}{3}}^2) \\ &= z_1^2 (1 + -1 + i\sqrt{3}) \\ &= z_1^2 (i\sqrt{3}) \\ &= z_1^2 \times \frac{z_2}{z_1} \\ &= z_1 z_2 \quad \text{--- 1} \end{aligned}$$

(ii) clearly $(z_1 - z_3)^2 + (z_2 - z_3)^2 = (z_1 - z_3)(z_2 - z_3)$

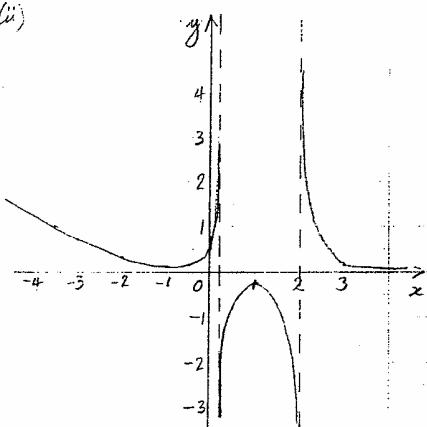
$$\begin{aligned} z_1^2 - 2z_1 z_3 + z_3^2 + z_2^2 - 2z_2 z_3 + z_3^2 &= \cancel{z_1^2} \cancel{z_2^2} \\ &= z_1 z_2 - z_1 z_3 - z_2 z_3 + z_3^2 \end{aligned}$$

$$\therefore \underline{z_1^2 + z_2^2 + z_3^2 = z_1 z_3 + z_2 z_3 + z_1 z_2} \quad \text{Q.E.D.} \quad (2)$$

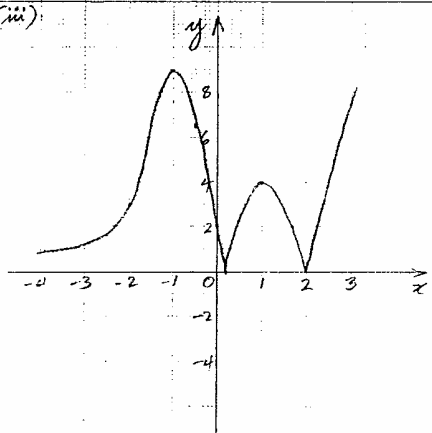
3(a)
(i)



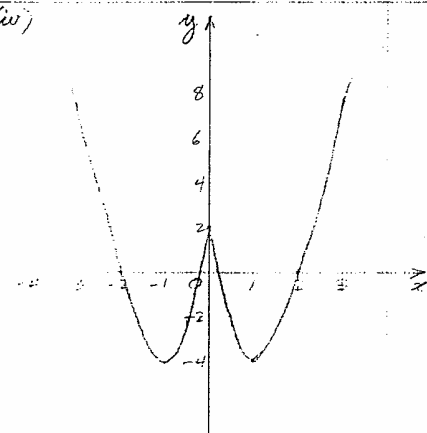
(ii)



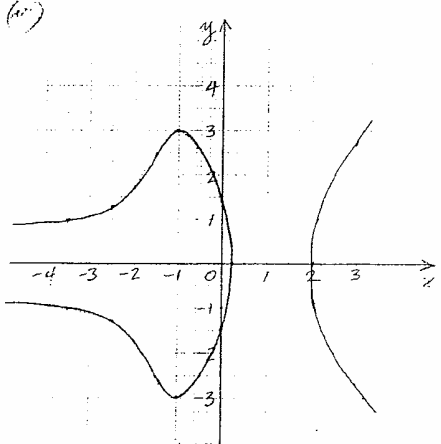
(iii)



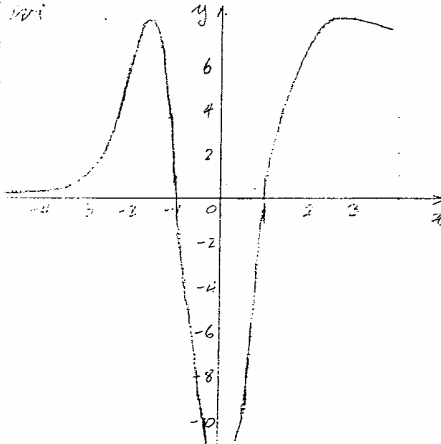
(iv)



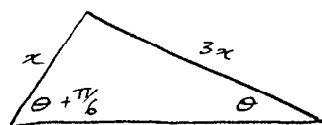
(v)



(vi)



3(b)



$$\frac{\sin \theta}{x} = \frac{\sin(\theta + \pi/6)}{3x} \quad \checkmark$$

$$3 \sin \theta = \sin \theta \cos \pi/6 + \cos \theta \sin \pi/6$$

$$3 \sin \theta = \frac{\sqrt{3}}{2} \sin \theta + \frac{1}{2} \cos \theta \quad \checkmark$$

$$6 \sin \theta - \sqrt{3} \sin \theta = \cos \theta$$

$$\frac{\sin \theta}{\cos \theta} \{6 - \sqrt{3}\} = 1$$

$$\tan \theta = \frac{1}{6 - \sqrt{3}}$$

$$\therefore \theta = \tan^{-1}\left(\frac{1}{6 - \sqrt{3}}\right) \quad \checkmark$$

4(a) (i) zeroes $1 \pm i, 3 \pm i$

$$[x - (1+i)][x - (1-i)] = x^2 - x + ix - x - ix + 1 + 1$$

$$= x^2 - 2x + 2 \quad \checkmark$$

$$[x - (3+i)][x - (3-i)] = x^2 - 6x + 10$$

$$\therefore P(x) = (x^2 - 2x + 2)(x^2 - 6x + 10)$$

(ii) Both factors are positive definite ($\Delta < 0$, coefft. of $x^2 > 0$) so their product must be positive.

4(b) (i) $P(x) = x^3 + 3px + q$

$$P'(x) = 3x^2 + 3p$$

$$P'(k) = 3k^2 + 3p = 0$$

$$\therefore p = -k^2$$

$$(ii) P(k) = k^3 + 3pk + q = 0$$

$$k = (-p)^{1/2}$$

$$\therefore -p \cdot (-p)^{1/2} + 3p(-p)^{1/2} + q = 0$$

$$\frac{q}{2p} = (-p)^{1/2}$$

$$\frac{q^2}{4p^2} = -p$$

$$4p^3 + q^2 = 0 \quad \checkmark$$

$$4(b) \text{ (iii) } x^3 + 3(-2i)x + (4-4i) = P(x)$$

Let zeroes be k, k, w

$$-k^2 = -2i$$

$$k = \sqrt{2i}$$

$$k = \pm(1+i) \quad \checkmark$$

$$k^2 w = -4 + 4i$$

$$w = \frac{-4+4i}{2i} \times \frac{2i}{2i}$$

$$= \frac{+8+8i}{+4}$$

$$= 2+2i$$

$$2k+w = \cancel{0}$$

$$2k = -2-2i$$

$$k = -1-i \quad \checkmark$$

$$\text{So } P(x) = (x+1+i)^2(x-2-2i).$$

$$(a+ib)^2 = 2i$$

$$a^2 - b^2 + 2aib = 2i$$

$$a^2 - b^2 = 0$$

$$a^2 + b^2 = 2$$

$$2a^2 = 2$$

$$a = \pm b = \pm 1$$

$$\text{but } ab = 1$$

$$\therefore a = b = \pm 1$$

$$4(c) \text{ (i) L.H.S.} = \left(x - \frac{3}{x}\right)^3 + 9\left(x - \frac{3}{x}\right) + 26$$

$$= x^3 - 3 \cdot x^2 \cdot \frac{3}{x} + 3 \cdot x \cdot \frac{9}{x^2} - \frac{27}{x^3} + 9x - \frac{27}{x} + 26$$

$$= x^3 + 26 - \frac{27}{x^3}$$

$$\text{R.H.S.} = \frac{x^6 + 26x^3 - 27}{x^3} \quad \checkmark$$

$$= \text{L.H.S.}$$

$$\text{(ii) } \frac{g(x^3)}{x^3} = \frac{(x^3+27)(x^3-1)}{x^3}$$

$$\text{Put } x = \rho - \frac{3}{\rho}$$

$$f(x) = 0 = \frac{(\rho^3+27)(\rho^3-1)}{\rho^3}$$

$$\therefore \rho = -3, 1 \quad \text{if } \rho \neq 0$$

$$x = -2 \quad \checkmark$$

$$f(x) = 0 = (x+2)(x^2-2x+13)$$

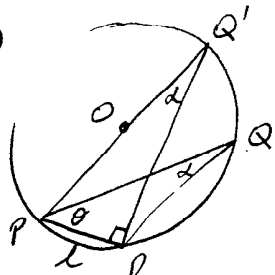
$$\therefore x = -2, 1 \pm 2\sqrt{3}i \quad \checkmark$$

$$\begin{array}{r} x^2 - 2x + 13 \\ x+2 \overline{) x^3 + 0x^2 + 9x + 26} \\ \underline{x^3 + 2x^2} \\ -2x^2 + 9x \\ \underline{-2x^2 - 4x} \\ 13x + 26 \end{array}$$

$$x = \frac{2 \pm \sqrt{4-52}}{2}$$

$$= 1 \pm 2\sqrt{3}i$$

4(d)(i)



$$\begin{aligned} \angle PQ'R &= \alpha \text{ (angle standing on same arc)} \\ \angle PRQ' &= \frac{\pi}{2} \text{ (angle in semicircle)} \\ PQ' &= 2r \text{ (diameter)} \\ \sin \alpha &= \frac{l}{2r} \end{aligned}$$

$$l = 2r \sin \alpha. \quad \checkmark$$

$$(ii) \frac{l}{\sin \alpha} = \frac{QR}{\sin \theta}$$

$$QR = \frac{l \sin \theta}{\sin \alpha}$$

$$\angle PRQ = \pi - \theta - \alpha$$

$$\text{Area} = \frac{1}{2} \cdot PR \cdot RQ \sin \angle PRQ$$

$$c^2 = 2c^2 - 1$$

$$c^2 = \frac{1+c^2}{2}$$

$$= \frac{1}{2} \cdot \frac{l \sin \theta}{\sin \alpha} \cdot \sin(\pi - \theta - \alpha) \quad \checkmark$$

$$= \frac{1}{2} \cdot \frac{l \sin \theta}{\sin \alpha} \cdot \sin(\theta + \alpha) \quad \checkmark$$

$$\begin{aligned} &= 2r^2 \sin \alpha \left\{ \sin \theta (\sin \theta \cos \alpha + \cos \theta \sin \alpha) \right\} \\ &= 2r^2 \sin \alpha \left\{ (1 - \cos^2 \theta) \cos \alpha + \sin \theta \cos \theta \sin \alpha \right\} \\ &= r^2 \sin \alpha \left\{ (2 - 1 - \cos 2\theta) \cos \alpha + \sin 2\theta \sin \alpha \right\} \\ &= r^2 \sin \alpha \left\{ \cos \alpha - \cos(2\theta + \alpha) \right\} \end{aligned}$$

(iii) If $PQ = QR$ then $2\theta + \alpha = \pi$ (isosceles Δ)

$$\text{Area} = r^2 \sin \alpha (\cos \alpha - \cos \pi)$$

$$= r^2 \sin \alpha (\cos \alpha + 1) \quad \checkmark$$

Question 5

- (a) (i) Taking the downward direction as positive we get the following force equation $m\ddot{x} = mg - mkv$, the resistance is negative as it OPPOSES the motion and is therefore directed upwards.
So $m\ddot{x} = mg - mkv \Rightarrow \ddot{x} = g - kv$

- (ii) The terminal velocity is when the net acceleration of the mass is 0 ie
 $\ddot{x} = g - kv = 0 \Rightarrow V_T = g/k$.

- (iii) Taking $\ddot{x} = \frac{dv}{dt} \Rightarrow \frac{dv}{dt} = g - kv$
 $\frac{dt}{dv} = \frac{1}{g - kv} = -\frac{1}{k} \left(\frac{-k}{g - kv} \right)$
 $t = -\frac{1}{k} \ln|g - kv| + c$
 $t = 0, v = 0 \Rightarrow c = \frac{1}{k} \ln g$
 $\therefore t = -\frac{1}{k} \ln \left| \frac{g - kv}{g} \right| \Rightarrow e^{-tk} = \left| \frac{g - kv}{g} \right| = \frac{g - kv}{g}$

We can remove the absolute value brackets since the initial direction is positive and it doesn't come to rest until it hits the ground so $\frac{g - kv}{g} > 0$

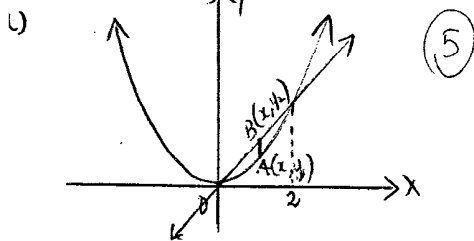
$$g - kv = ge^{-tk} \Rightarrow v = \frac{g}{k}(1 - e^{-tk})$$

- (iv) $v = \frac{g}{k}(1 - e^{-tk}) \Rightarrow x = \int \frac{g}{k}(1 - e^{-tk}) dt$
 $x = \frac{g}{k} \left(t + \frac{1}{k} e^{-tk} \right) + c_1$
 $t = 0, x = 0 \Rightarrow c_1 = -\frac{g}{k^2}$
 $\therefore x = \frac{g}{k} \left(t + \frac{1}{k} e^{-tk} - \frac{1}{k} \right)$

- (b) (i) If we choose 5 players to form a team, this can be done in $\binom{10}{5}$ ways. But the remaining 5 will form the opposing team. So there are $\frac{1}{2} \binom{10}{5} = 126$
- (ii) If the twins are on one team then the remaining team can be formed in $\binom{8}{3} = 56$ ways. So the probability that the twins are on the same team is $56/126 = 4/9$.

- (5) (c) (i) There are $4! = 24$ ways to arrange everyone without restrictions. However with 3 men and only 2 women there must be one pair of men sitting next to each other.
Probability = 0.
- (ii) There are $4!$ ways to sit everyone down without restriction. Seat two women down next to each other. This leaves $2! = 2$ ways to seat the men down and 1 way to sit the other woman.
The two women can be chosen in $\binom{3}{2} = 3$ ways. Then the two women seated together can swap seats
So there are $2 \times 3 \times 2 = 12$ ways. to sit everyone down.
So the probability is $12/24 = 1/2$.
- (iii) There are $(2n!)$ ways to sit everyone down without restrictions. Choose two women first and sit them down. This can be done in $\binom{n+1}{2} = \frac{n(n+1)}{2}$ ways. The remaining women can be seated in $(n-1)!$ ways. Then the men can be seated in $n!$ ways.
The two women together can be swapped around.
A total of $\frac{n(n+1)}{2} \times (n-1)! \times n \times 2 = (n+1)(n!)^2$.
So the probability is $\frac{(n+1)(n!)^2}{(2n!)} = \frac{(n+1)n!}{(2n!)}$.

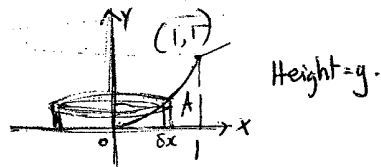
QUESTION 6



$$\begin{aligned}
 x^2 = 2x &\Rightarrow x=0 \text{ or } 2 \\
 \text{Diameter} &= y_2 - y_1 = 2x - x^2 \\
 A &= \text{area of semi-circle} = \frac{\pi}{2} \left(x - \frac{x^2}{2}\right)^2 \\
 \text{Thickness of solid} &= \delta x \\
 \text{Vol. of element} &= \delta V = A \delta x \\
 &= \frac{\pi}{2} \left(x - \frac{x^2}{2}\right)^2 \delta x \\
 \text{Total Volume} &= \lim_{\delta x \rightarrow 0} \sum_{x=0}^2 \frac{\pi}{2} \left(x - \frac{x^2}{2}\right)^2 \delta x \\
 &= \frac{\pi}{2} \int_0^2 \left(x - \frac{x^2}{2}\right)^2 dx \\
 &= \frac{\pi}{2} \int_0^2 \left(x^2 - x^3 + \frac{x^4}{4}\right) dx \\
 &= \frac{\pi}{2} \left[\frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{20} \right]_0^2 \\
 &= \frac{\pi}{2} \left[\frac{8}{3} - 4 + \frac{32}{20} - 0 \right] \\
 &= \frac{2\pi}{15} \text{ units}^3
 \end{aligned}$$

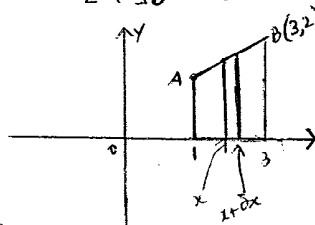
$$\begin{aligned}
 \text{ii) } l &= \int_0^4 \sqrt{1 + \left(\frac{3}{2}x\right)^2} dx \\
 &= \int_0^4 \sqrt{1 + \frac{9}{4}x} dx \\
 &= \left[\frac{\left(\frac{9}{4}x + 1\right)^{3/2}}{\frac{3}{2} \cdot \frac{9}{4}} \right]_0^4 \\
 &= \frac{8}{27} \left[10^{3/2} - 1^{3/2} \right] \\
 &= \frac{8}{27} [10\sqrt{10} - 1] \text{ units}^3
 \end{aligned}$$

(c)



$$\begin{aligned}
 \text{Vol. when region A is rotated about y axis} \\
 \text{Vol. of shell } \delta V &= \pi [(x + \delta x)^2 - x^2] y \\
 &= \pi [x^2 + 2x\delta x + \delta x^2 - x^2] y \\
 &= 2\pi xy \delta x
 \end{aligned}$$

$$\begin{aligned}
 V &= 2\pi \int_0^1 x \cdot x^2 dx \\
 &= 2\pi \left[\frac{x^4}{4} \right]_0^1 = \frac{\pi}{2} \text{ units}^3
 \end{aligned}$$



$$\begin{aligned}
 \text{Eqn of line AB is } x &= 2y - 1 \\
 \text{or } y &= \frac{x+1}{2}
 \end{aligned}$$

$$\begin{aligned}
 V &= 2\pi \int_1^3 x \left(\frac{x+1}{2}\right) dx \\
 &= \pi \int_1^3 (x^2 + x) dx
 \end{aligned}$$

$$\begin{aligned}
 &= \pi \left[\frac{x^3}{3} + \frac{x^2}{2} \right]_1^3 \\
 &= \pi \left[\left(9 + \frac{9}{2}\right) - \left(\frac{1}{3} + \frac{1}{2}\right) \right] = \frac{38\pi}{3}
 \end{aligned}$$

$$\therefore \text{Total volume} = \frac{\pi}{2} + \frac{38\pi}{3} = \frac{79\pi}{6}$$

$$\text{Vol. of bowl} = 18\pi - \frac{79\pi}{6} = \frac{29\pi}{6} \text{ units}^3$$

Question 7

$$(a) \quad \left(\sqrt{x} + \frac{a}{x} \right)^{10} = \left(x^{\frac{1}{2}} + ax^{-1} \right)^{10} = \sum_{r=0}^{10} {}^{10}C_r \left(x^{\frac{1}{2}} \right)^{10-r} (ax^{-1})^r = \sum_{r=0}^{10} (a^r \times {}^{10}C_r) x^{5-3r/2}$$

To get the coefficient of x^{-4} we need $5 - 3r/2 = -4 \Rightarrow -3r/2 = -9 \Rightarrow r = 6$

So the coefficient of $x^{-4} = a^6 \times {}^{10}C_6 = 13440$

$$a^6 = \frac{13440}{{}^{10}C_6} = 64 \Rightarrow a = 2$$

- (b) (i) $CP^2 = AC \times BC$ by the rule for intercepts and tangents.
Similarly $CQ^2 = AC \times BC \Rightarrow CP^2 = CQ^2 \Rightarrow CP = CQ$ as $(CP, CQ > 0)$
- (ii) ΔPQC is isosceles with $\angle CPQ = \angle CQP = x$ (base angles of isos. Δ)
 $\angle BAP = \angle CPQ = x$ (alternate segment theorem). Similarly $\angle BAQ = x$
So $\angle CAQ = \angle CPQ$ both of which stand on chord CQ .
So A, P, C & Q are concyclic points by the converse of angles in the same segment theorem.
- (iii) The exterior angle $\angle DAP = 180 - 2x$, DAQ is a straight line.
So $\angle DAC = 180 - x$. Now $BADP$ is a cyclic quad so that
 $\angle BPD + \angle BAD = 180^\circ \Rightarrow \angle BAD = x$.
Thus $\angle CPB = \angle BPD = x \Rightarrow PB$ bisects $\angle CPD$.

(c)

$$\begin{aligned} (1) \quad & T(k, x) - T(k, x+1) \\ &= \frac{k!}{x(x+1) \cdots (x+k)} - \frac{k!}{(x+1)(x+2) \cdots (x+k)(x+k+1)} \\ &= \frac{k!(x+k+1) - k!x}{x(x+1) \cdots (x+k)(x+k+1)} \\ &= \frac{(k+1)!}{x(x+1) \cdots (x+k+1)} = T(k+1, x) \end{aligned}$$

7 (ii) Test $n = 1$:

$$\text{LHS} = T(1, x) = \frac{{}^1C_0}{x} - \frac{{}^1C_1}{x+1} = \frac{1}{x} - \frac{1}{x+1} = \frac{1}{x(x+1)} = \frac{1!}{x(x+1)} = \text{RHS}$$

So the statement is true for $n = 1$

Assume true for some integer $n = k$.

$$\text{ie } T(k, x) = \frac{k!}{x(x+1)(x+2)\cdots(x+k)}$$

We need to prove the statement is true for $n = k + 1$

$$\text{ie } T(k+1, x) = \frac{(k+1)!}{x(x+1)(x+2)\cdots(x+k)(x+k+1)}$$

$$\text{LHS} = T(k+1, x)$$

$$= T(k, x) - T(k, x+1)$$

from (i)

$$= \frac{(k+1)!}{x(x+1)(x+2)\cdots(x+k)(x+k+1)}$$

$$= \text{RHS}$$

(iii) Substitute $x = 1/2$ into both sides of the result from (ii) and simplify

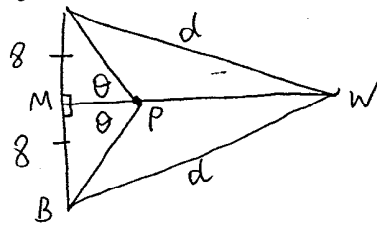
$$\begin{aligned} \frac{{}^nC_0}{\frac{1}{2}} - \frac{{}^nC_1}{\frac{1}{2}+1} + \frac{{}^nC_2}{\frac{1}{2}+2} - \dots + (-1)^n \frac{{}^nC_n}{\frac{1}{2}+n} &= \frac{n!}{\frac{1}{2} \cdot \frac{3}{2} \cdot \dots \cdot (n+\frac{1}{2})} \\ &= \frac{2^{n+1} n!}{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n+1)} \\ \frac{{}^nC_0}{1} - \frac{{}^nC_1}{3} + \frac{{}^nC_2}{5} - \dots + (-1)^n \frac{{}^nC_n}{2n+1} &= \frac{2^n n!}{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n+1)} \end{aligned}$$

8.

$$\begin{aligned}
 (a) \quad V &= \pi \int_0^{\pi} (c - \sin x)^2 dx & 2 \\
 &= \pi \int_0^{\pi} c^2 - 2c \sin x + \frac{1}{2} - \frac{1}{2} \cos 2x dx \\
 &= \pi \left[c^2 x + 2c \cos x + \frac{1}{2} x - \frac{1}{4} \sin 2x \right]_0^{\pi} \\
 &= \pi \left[c^2 \pi - 2c + \frac{1}{2} \pi - 0 - (0 + 2c + 0 - 0) \right] \\
 &= \pi \left[c^2 \pi - 4c + \frac{1}{2} \pi \right] & 1 \\
 V' &= \pi(2c\pi - 4) \\
 V'' &= \pi(2\pi) > 0 & 1 \\
 \text{Min occurs when } V' &= 0 & (5) \\
 \text{ie. } 2c\pi &= 4 \\
 c &= \frac{2}{\pi}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad f(\theta) &= \frac{2 - \cos \theta}{\sin \theta}, \quad 0 < \theta < \frac{\pi}{2} \\
 f'(\theta) &= \frac{1 - 2\cos \theta}{\sin^2 \theta} \quad (\text{by quotient rule}) \\
 \text{Start pt where } f'(\theta) &= 0 \\
 1 - 2\cos \theta &= 0 \\
 \cos \theta &= \frac{1}{2} & \theta = \frac{\pi}{3} & 3 \\
 \text{when } \theta = 1, \quad f'(\theta) &> 0 & \therefore \text{Min value} \\
 \theta = 1.2, \quad f'(\theta) &< 0 & \text{when } \theta = \frac{\pi}{3} \\
 f\left(\frac{\pi}{3}\right) &= \frac{2 - \cos \frac{\pi}{3}}{\sin \frac{\pi}{3}} = \frac{2 - \frac{1}{2}}{\frac{\sqrt{3}}{2}} = \sqrt{3}
 \end{aligned}$$

b (ii) A



1 Pythag

+ 1 cosec or cot

3 correct.

$$PW = \sqrt{d^2 - 64} - 8 \cot \theta$$

$$L = 2PA + PW$$

$$= 2 \times 8 \operatorname{cosec} \theta + \sqrt{d^2 - 64} - 8 \cot \theta$$

$$= 16 \cdot \frac{1}{\sin \theta} - 8 \frac{\cos \theta}{\sin \theta} + \sqrt{d^2 - 64}$$

$$= 8 \left(\frac{2 - \cos \theta}{\sin \theta} \right) + \sqrt{d^2 - 64} = 8f(\theta) + \sqrt{d^2 - 64}$$

Clearly $\sin \theta \leq 1$

$\sin \theta = \frac{8}{AP}$. Now $d = AW \geq AP$ (triangle inequality)

$$\text{So } \sin \theta = \frac{8}{AP} \geq \frac{8}{d}$$

$$\text{ie } \frac{8}{d} \leq \sin \theta \leq 1$$

$$\text{iii) If } d = 20 \quad L = 8f(\theta) + \sqrt{336}$$

$$L' = 8f'(\theta)$$

Min occurs when $f'(\theta) = 0$

by part (ii) min value of L is

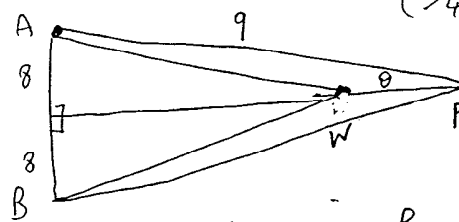
$$L = 8\sqrt{3} + \sqrt{336}$$

when min, length of MP is $8 \cot \frac{\pi}{3} = \frac{8}{\sqrt{3}}$

We can use this value of θ because $\frac{8}{20} \leq \sin \theta \leq 1 \Rightarrow \sin^{-1}(2/5) \leq \theta \leq \pi/2$

and clearly $\theta = \pi/3$ satisfies this inequality.

(iv) Note $MW = \sqrt{81-64} = \sqrt{17} \begin{matrix} > 4 \\ < \frac{8}{\sqrt{3}} \end{matrix}$



Key idea is P is beyond W.

2

More appropriate alternative solution:

We can't use the same method because with $d = 9$ we get

$\frac{8}{9} \leq \sin \theta \leq 1 \Rightarrow \sin^{-1}(8/9) \leq \theta \leq \pi/2$ and clearly $\theta = \pi/3$ does NOT satisfy this inequality.

So in this range we have a 1:1 function for L (you can quickly show that it is increasing) so all we need to is test the end points of $\sin^{-1}(8/9) \leq \theta \leq \pi/2$ ie substitute $\theta = \sin^{-1}(8/9)$ and $\theta = \pi/2$ into the formula for L and take the minimum value of L resulting.